

Introduction to Computer Vision



Lec.06: Image stitching
2022.10.24

Panorama



360 VR



360 VR

360 Camera
GoPro odyssey



Image stitching



VR Glasses
Huawei



How to combine two images?



How to combine two images?



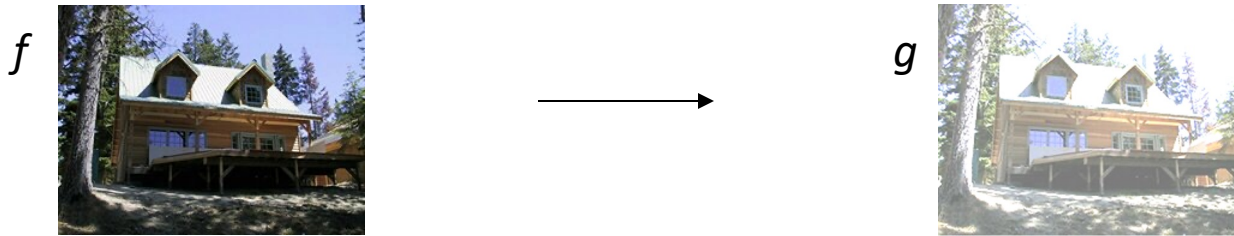
What is image warping?
How to compute it?

Today

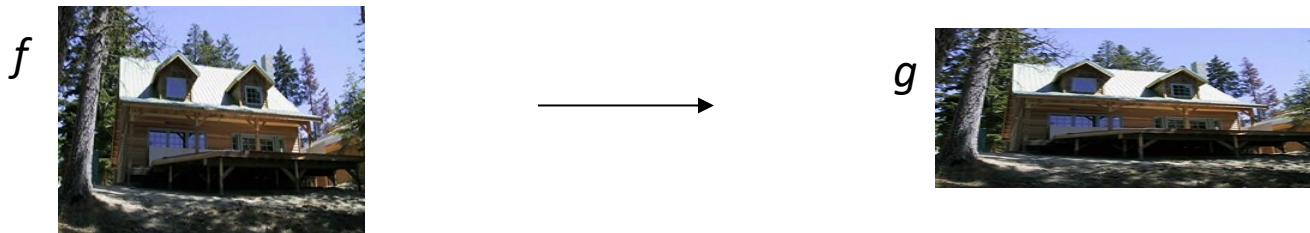
- **Image Warping**
- Image Stitching

Image Warping

- **image filtering**: change *intensity* of image



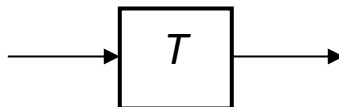
- **image warping**: change *shape* of image



Parametric (global) warping



$\mathbf{p} = (x, y)$



$\mathbf{p}' = (x', y')$

- Transformation T is a coordinate transformation:

$$\mathbf{p}' = T(\mathbf{p})$$

- Examples:



translation

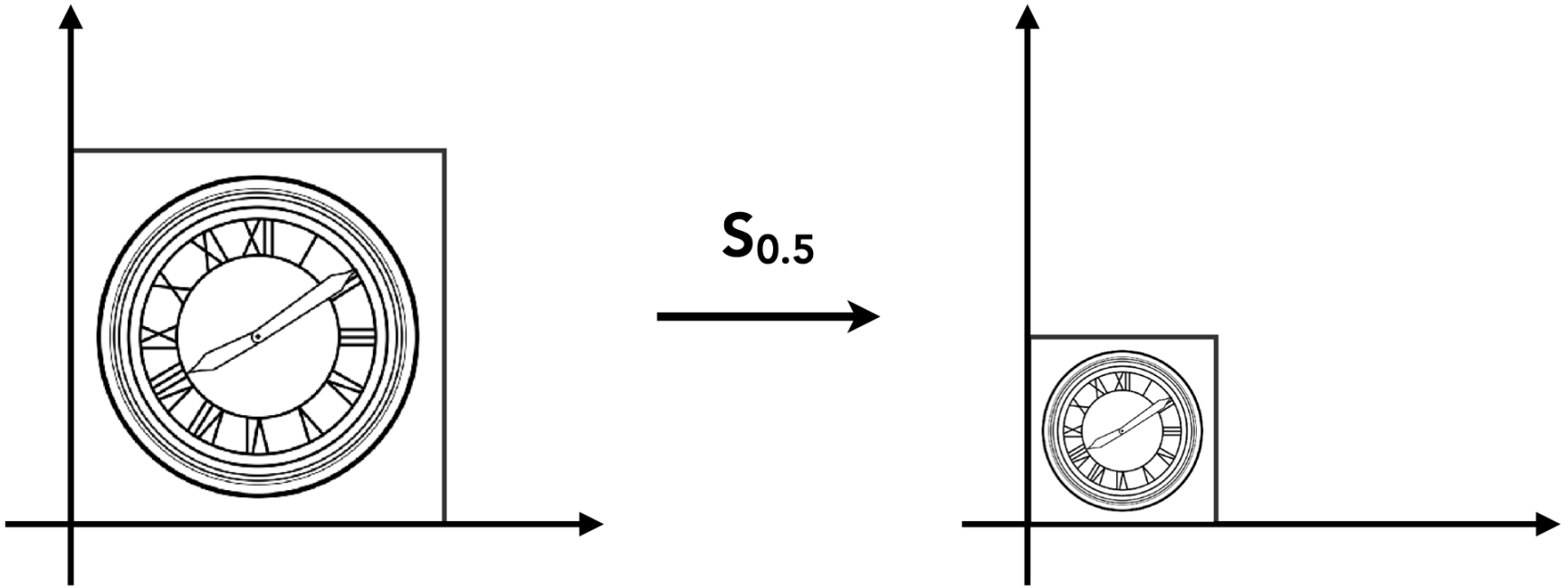


rotation



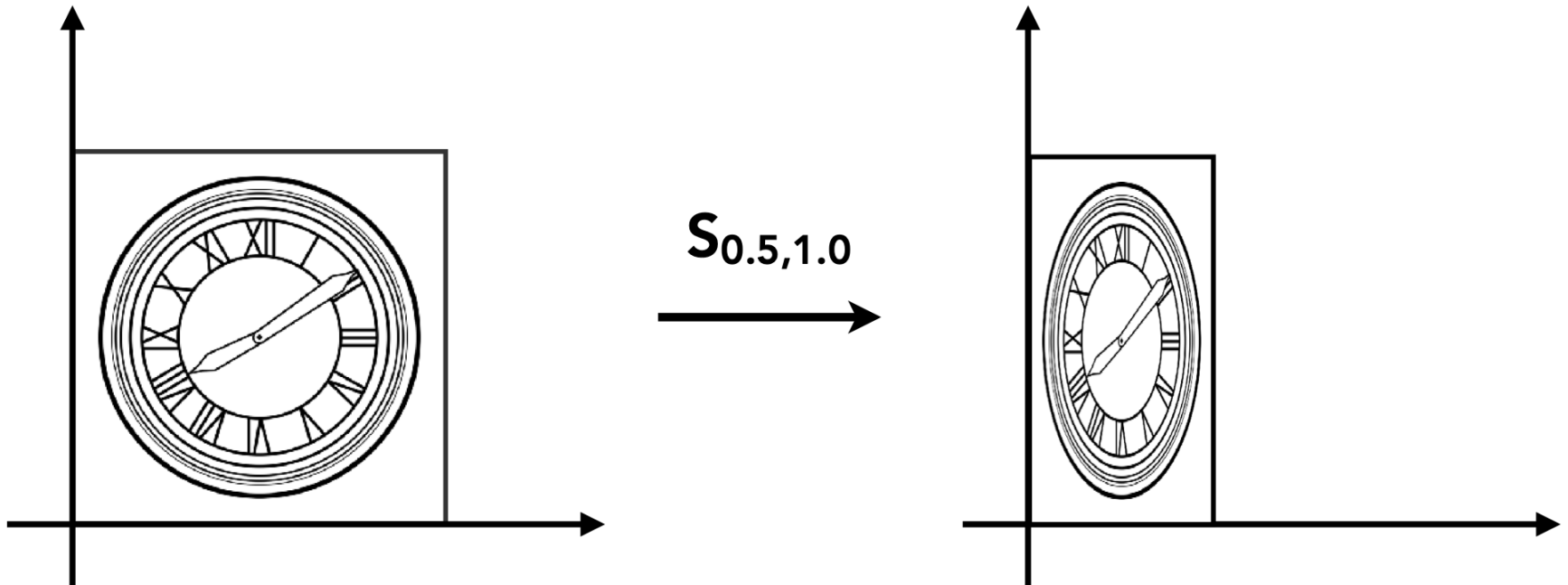
aspect

Scale Matrix



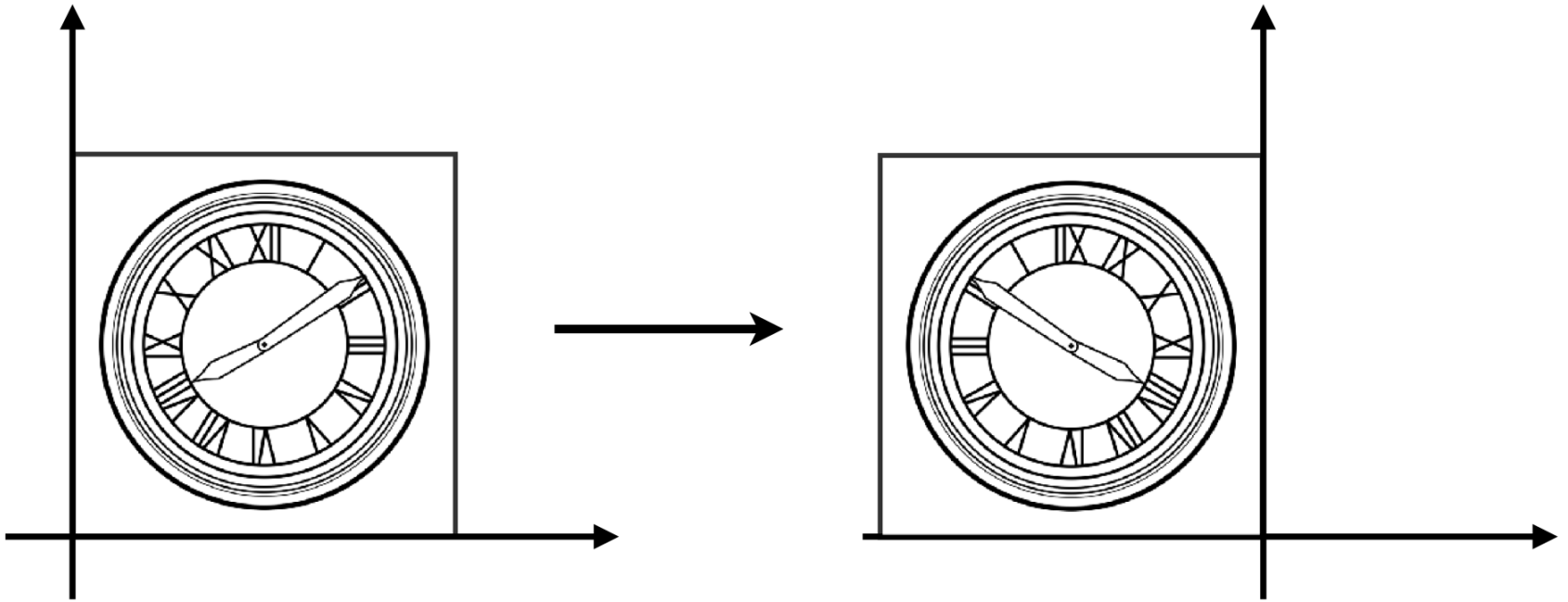
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Scale (Non-Uniform)



$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Reflection Matrix



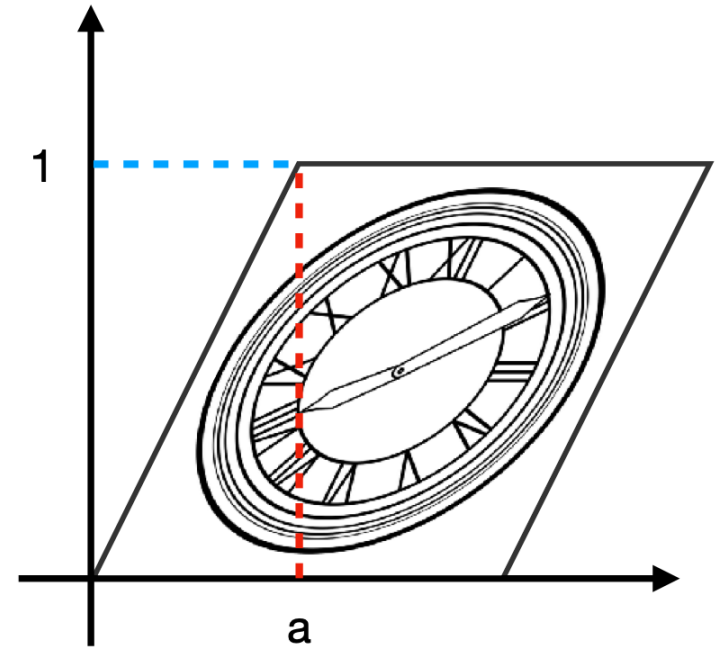
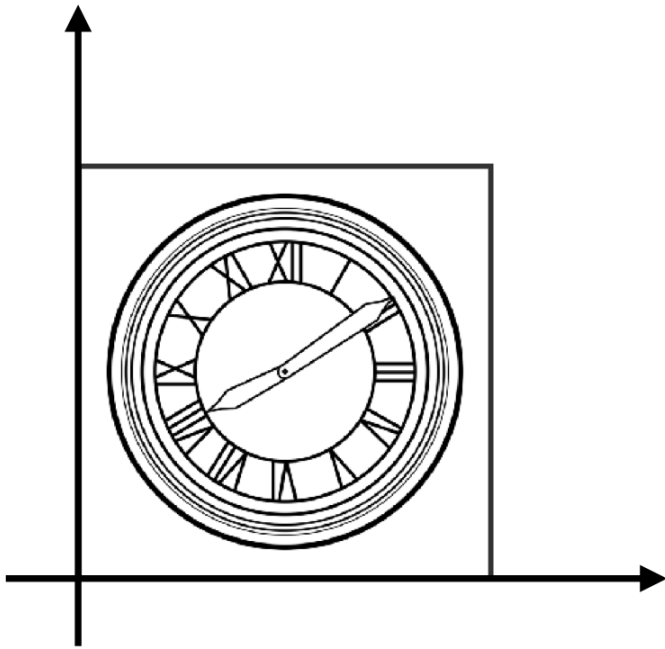
Horizontal reflection:

$$x' = -x$$

$$y' = y$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Shear Matrix



Hints:

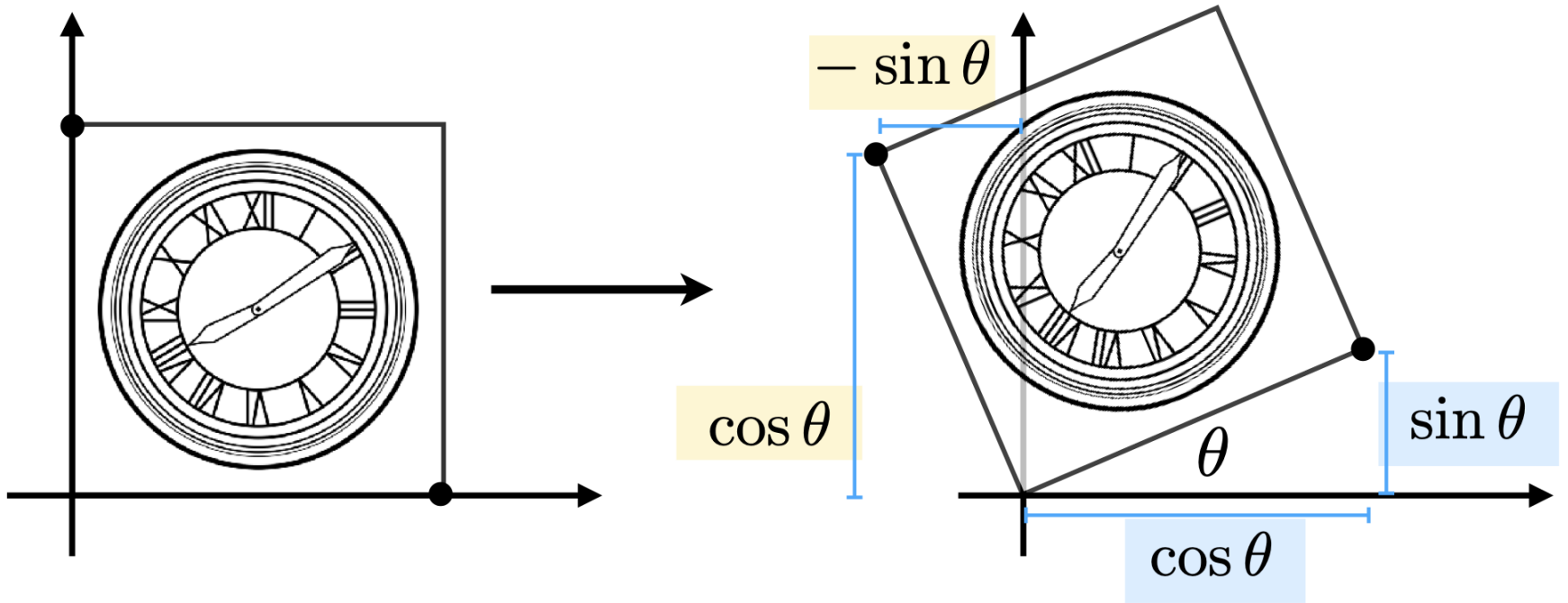
Horizontal shift is 0 at $y=0$

Horizontal shift is a at $y=1$

Vertical shift is always 0

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Rotation Matrix



$$\mathbf{R}_\theta = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

Linear Transforms = Matrices

$$x' = a x + b y$$

$$y' = c x + d y$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\mathbf{x}' = \mathbf{M} \mathbf{x}$$

Columns of $\mathbf{M}$ are new coordinates of the standard basis after transformation

Affine Transformations

Affine map = linear map + translation

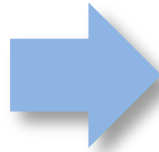
$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} t_x \\ t_y \end{pmatrix}$$

Using homogenous coordinates:

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{pmatrix} a & b & t_x \\ c & d & t_y \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

any transformation represented by a 3x3 matrix with last row $[0 \ 0 \ 1]$ we call an *affine* transformation

Projective Transformation (**Homography**)



Projective Transformation (Homography)

$$\begin{bmatrix} x'_i \\ y'_i \\ 1 \end{bmatrix} \cong \begin{bmatrix} h_{00} & h_{01} & h_{02} \\ h_{10} & h_{11} & h_{12} \\ h_{20} & h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix}$$

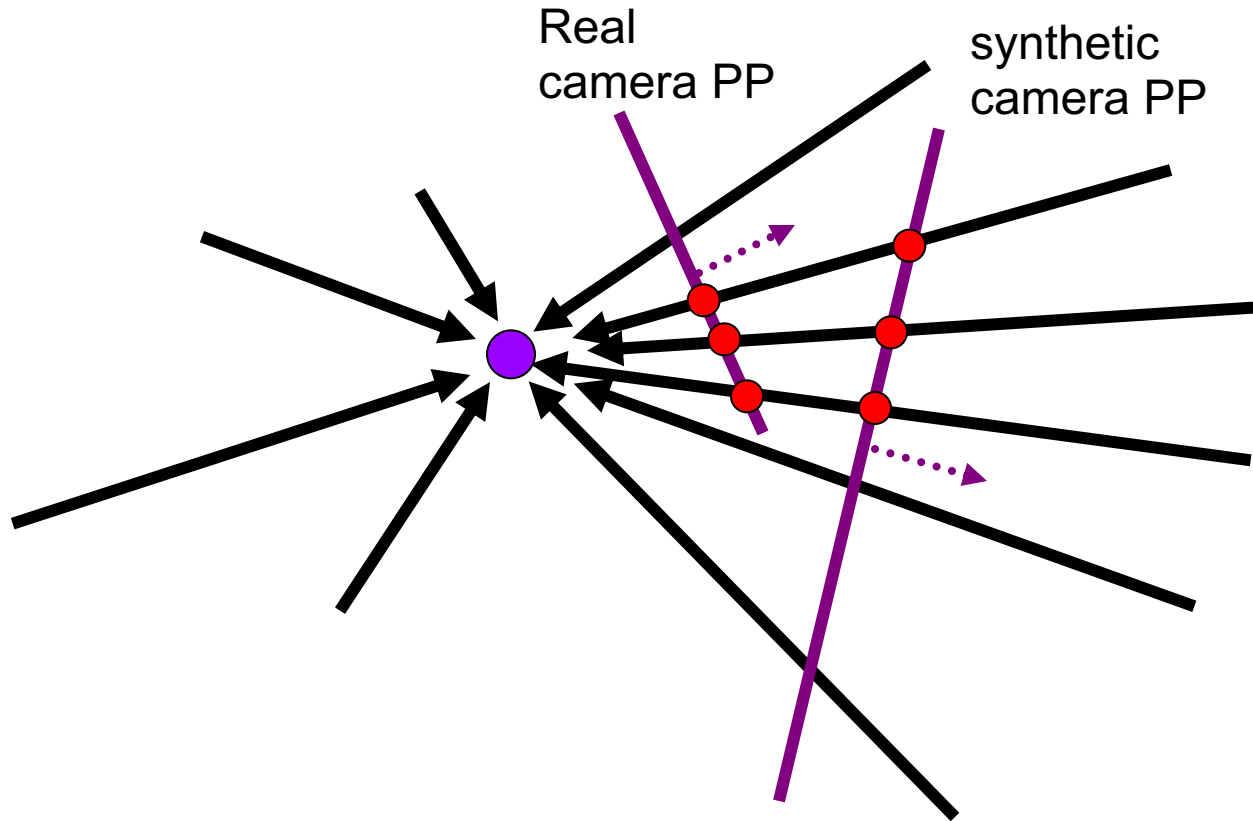
Maybe nonzero

$$x'_i = \frac{h_{00}x_i + h_{01}y_i + h_{02}}{h_{20}x_i + h_{21}y_i + h_{22}}$$

$$y'_i = \frac{h_{10}x_i + h_{11}y_i + h_{12}}{h_{20}x_i + h_{21}y_i + h_{22}}$$

- Homography matrix is up to scale (can be multiplied by a scalar), which means the degree of freedom is 8
- We usually constrain the length of the vector $[h_{00} \ h_{01} \dots \ h_{22}]$ to be 1

Change projection plane (pp)



Can generate any synthetic camera view
as long as it has **the same center of projection!**

Change projection plane (pp)

Original image



St.Petersburg
photo by A. Tikhonov

Virtual camera rotations

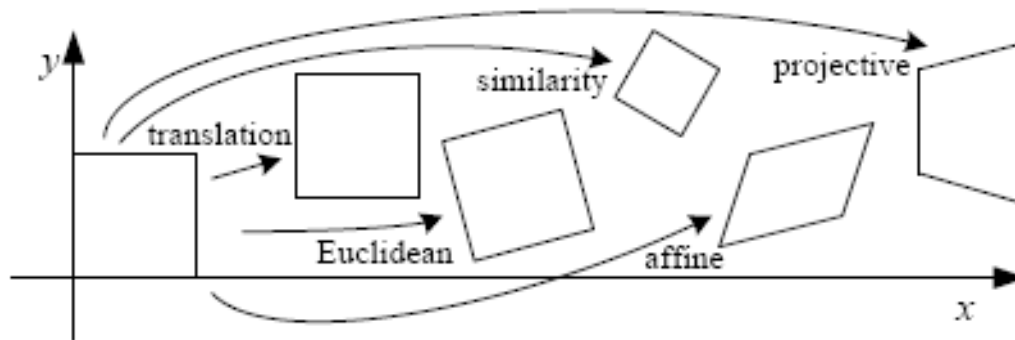


Projective Transformation (**Homography**)

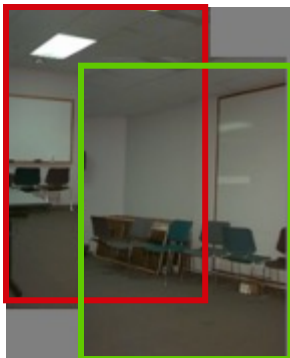
In what cases is the transformation of two images a homography?

- Camera rotated with its center unmoved
- Camera center moved and the scene is a plane

Summary of 2D transformations

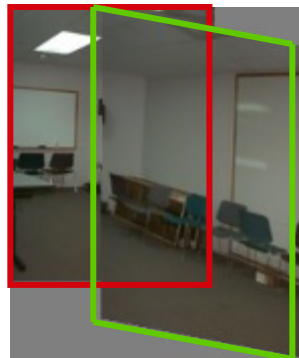


Translation



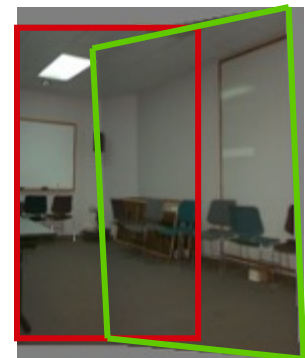
2 unknowns

Affine



6 unknowns

Projective

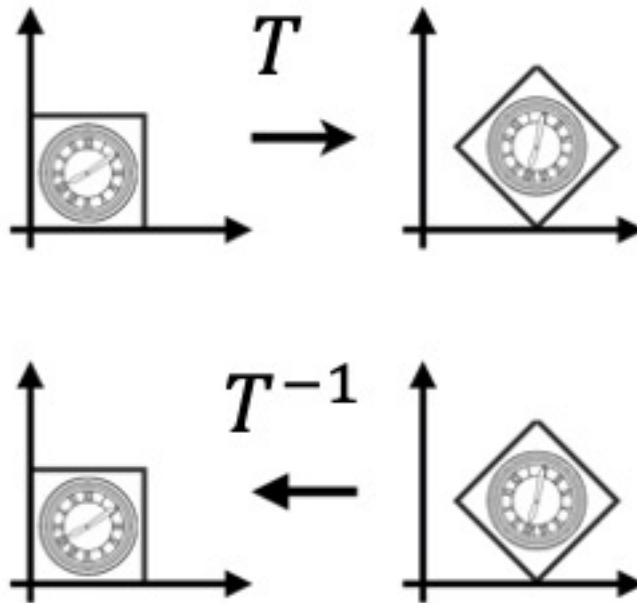


8 unknowns

Inverse Transform

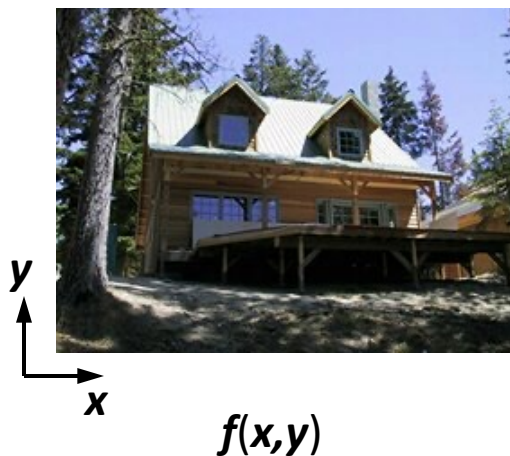
$$T^{-1}$$

T^{-1} is the inverse of transform T in both a matrix and geometric sense



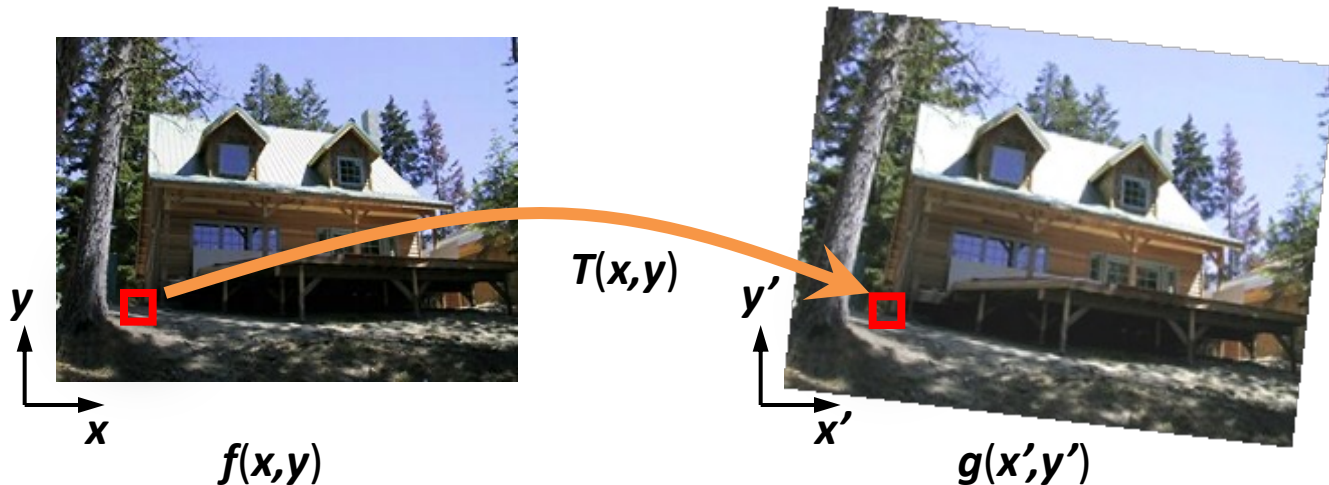
Implementing image warping

- Given a coordinate transform $(x',y') = T(x,y)$ and a source image $f(x,y)$, how do we compute an transformed image $g(x',y') = f(T(x,y))$?



Forward Warping

- Send each pixel $f(\mathbf{x})$ to its corresponding location $(\mathbf{x}', \mathbf{y}') = T(\mathbf{x}, \mathbf{y})$ in $g(\mathbf{x}', \mathbf{y}')$



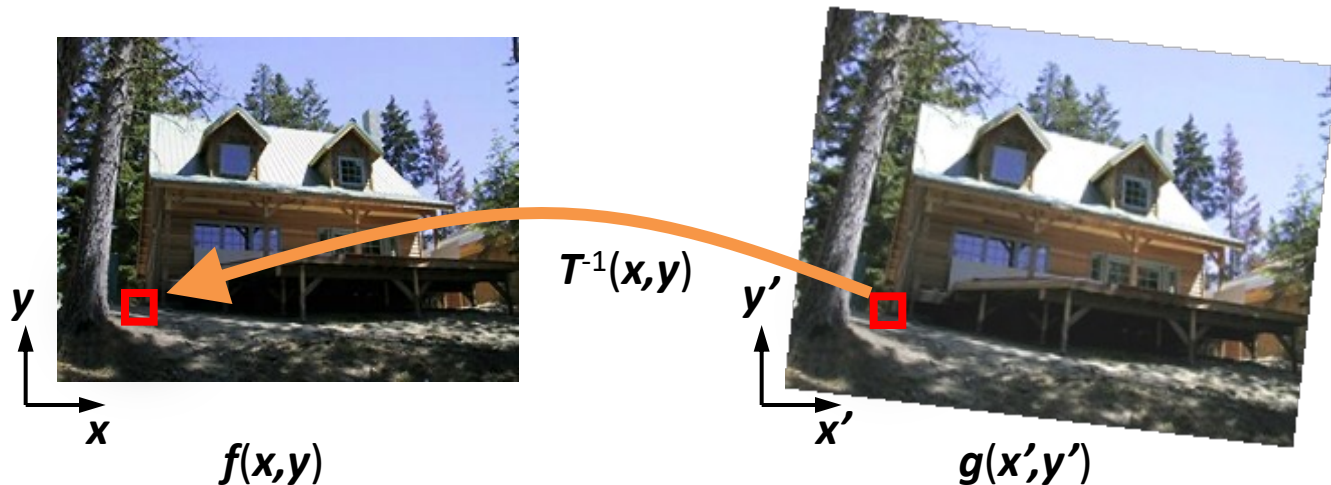
Forward Warping

- What if pixel lands “between” pixels?



Inverse Warping

- Get each pixel $g(x',y')$ from its corresponding location $(x,y) = T^{-1}(x',y')$ in $f(x,y)$



Inverse Warping

- What if pixel lands between pixels?

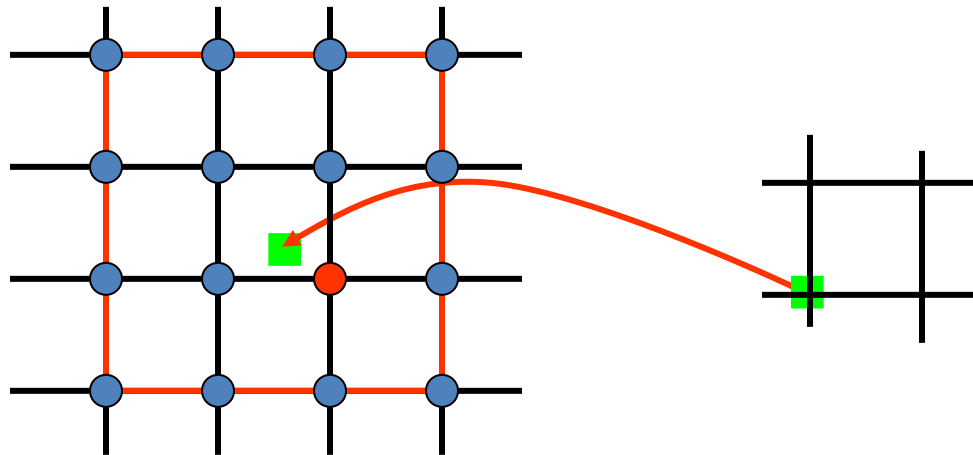


Answer: interpolate color values from neighboring pixels

Interpolation

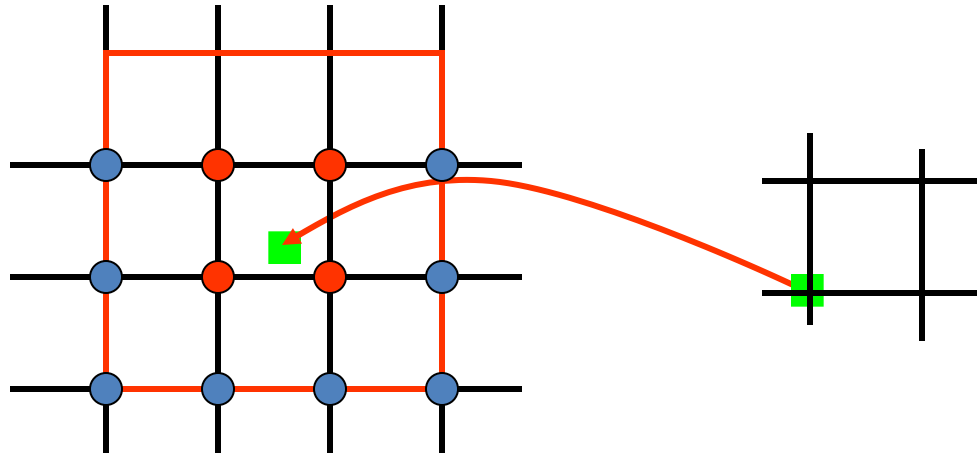
Nearest neighbor

- Copies the color of the pixel with the closest integer coordinate



Interpolation

Weighted sum of four neighboring pixels



Interpolation

- Possible interpolation filters:
 - nearest neighbor
 - bilinear
 - bicubic
 - sinc



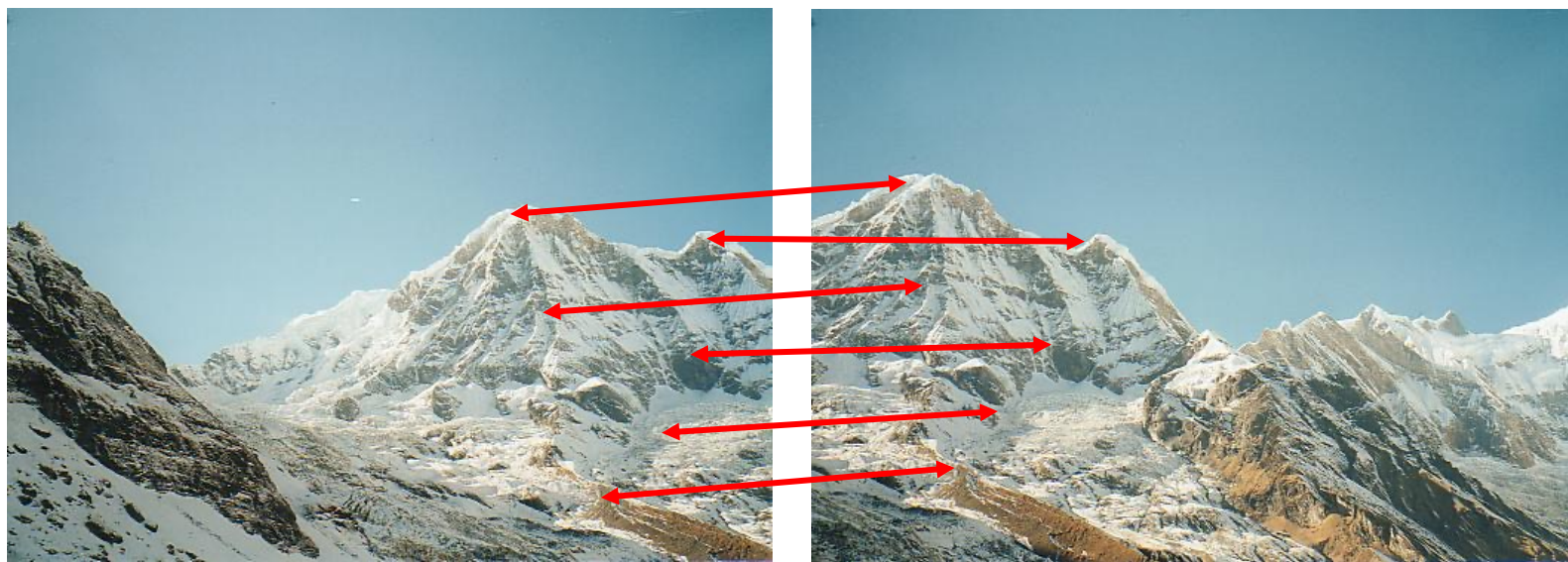
Questions?

Today

- Image Warping.
- **Image Stitching.**

How to compute transformation?

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} \cong T \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$



1. Image matching (each match gives an equation)
2. Solve T from the obtained matches

Affine transformations

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} ax + by + c \\ dx + ey + f \\ 1 \end{bmatrix}$$

- How many unknowns?
- How many equations per match?
- How many matches do we need?

Affine transformations

- For each match, we have

$$\begin{aligned}x' &= ax + by + c \\ y' &= dx + ey + f\end{aligned}$$

- Matrix form

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x & y & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & x & y & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \\ e \\ f \end{bmatrix}$$

Affine transformations

- For n matches

$$\begin{bmatrix}
 x_1 & y_1 & 1 & 0 & 0 & 0 \\
 0 & 0 & 0 & x_1 & y_1 & 1 \\
 x_2 & y_2 & 1 & 0 & 0 & 0 \\
 0 & 0 & 0 & x_2 & y_2 & 1 \\
 \vdots & & & & & \\
 x_n & y_n & 1 & 0 & 0 & 0 \\
 0 & 0 & 0 & x_n & y_n & 1
 \end{bmatrix}
 \begin{bmatrix}
 a \\
 b \\
 c \\
 d \\
 e \\
 f
 \end{bmatrix}
 =
 \begin{bmatrix}
 x'_1 \\
 y'_1 \\
 x'_2 \\
 y'_2 \\
 \vdots \\
 x'_n \\
 y'_n
 \end{bmatrix}$$

$$\mathbf{A}_{2n \times 6} \mathbf{t}_{6 \times 1} = \mathbf{b}_{2n \times 1}$$

How to solve $\mathbf{t}$?

- Least squares: find $\mathbf{t}$ that minimizes

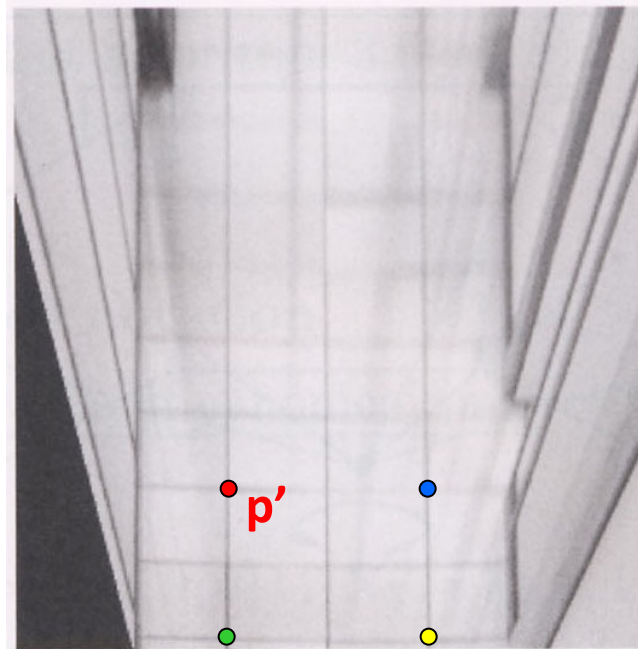
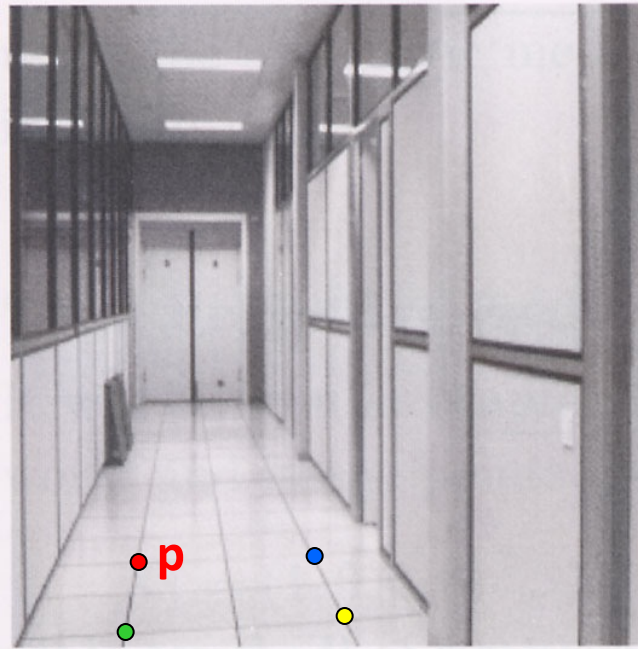
$$||\mathbf{A}\mathbf{t} - \mathbf{b}||^2$$

- The solution is given by

$$\mathbf{A}^T \mathbf{A} \mathbf{t} = \mathbf{A}^T \mathbf{b}$$

$$\mathbf{t} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b}$$

Projective transformations



$$\begin{bmatrix} x'_i \\ y'_i \\ 1 \end{bmatrix} \cong \begin{bmatrix} h_{00} & h_{01} & h_{02} \\ h_{10} & h_{11} & h_{12} \\ h_{20} & h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix}$$

p'

Homography

p

Solving for homographies

$$\begin{bmatrix} x'_i \\ y'_i \\ 1 \end{bmatrix} \cong \begin{bmatrix} h_{00} & h_{01} & h_{02} \\ h_{10} & h_{11} & h_{12} \\ h_{20} & h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix}$$

$$x'_i = \frac{h_{00}x_i + h_{01}y_i + h_{02}}{h_{20}x_i + h_{21}y_i + h_{22}}$$

$$y'_i = \frac{h_{10}x_i + h_{11}y_i + h_{12}}{h_{20}x_i + h_{21}y_i + h_{22}}$$

$$x'_i (h_{20}x_i + h_{21}y_i + h_{22}) = h_{00}x_i + h_{01}y_i + h_{02}$$

$$y'_i (h_{20}x_i + h_{21}y_i + h_{22}) = h_{10}x_i + h_{11}y_i + h_{12}$$

Solving for homographies

$$x'_i (h_{20}x_i + h_{21}y_i + h_{22}) = h_{00}x_i + h_{01}y_i + h_{02}$$

$$y'_i (h_{20}x_i + h_{21}y_i + h_{22}) = h_{10}x_i + h_{11}y_i + h_{12}$$

$$\begin{bmatrix} x_i & y_i & 1 & 0 & 0 & 0 & -x'_i x_i & -x'_i y_i & -x'_i \\ 0 & 0 & 0 & x_i & y_i & 1 & -y'_i x_i & -y'_i y_i & -y'_i \end{bmatrix} \begin{bmatrix} h_{00} \\ h_{01} \\ h_{02} \\ h_{10} \\ h_{11} \\ h_{12} \\ h_{20} \\ h_{21} \\ h_{22} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Solving for homographies

$$\begin{bmatrix}
 x_1 & y_1 & 1 & 0 & 0 & 0 & -x'_1 x_1 & -x'_1 y_1 & -x'_1 \\
 0 & 0 & 0 & x_1 & y_1 & 1 & -y'_1 x_1 & -y'_1 y_1 & -y'_1 \\
 & & & & & & \vdots & & \\
 x_n & y_n & 1 & 0 & 0 & 0 & -x'_n x_n & -x'_n y_n & -x'_n \\
 0 & 0 & 0 & x_n & y_n & 1 & -y'_n x_n & -y'_n y_n & -y'_n
 \end{bmatrix}
 \begin{bmatrix}
 h_{00} \\
 h_{01} \\
 h_{02} \\
 h_{10} \\
 h_{11} \\
 h_{12} \\
 h_{20} \\
 h_{21} \\
 h_{22}
 \end{bmatrix}
 =
 \begin{bmatrix}
 0 \\
 0 \\
 \vdots \\
 0 \\
 0
 \end{bmatrix}$$

$\mathbf{A}$
 $2n \times 9$

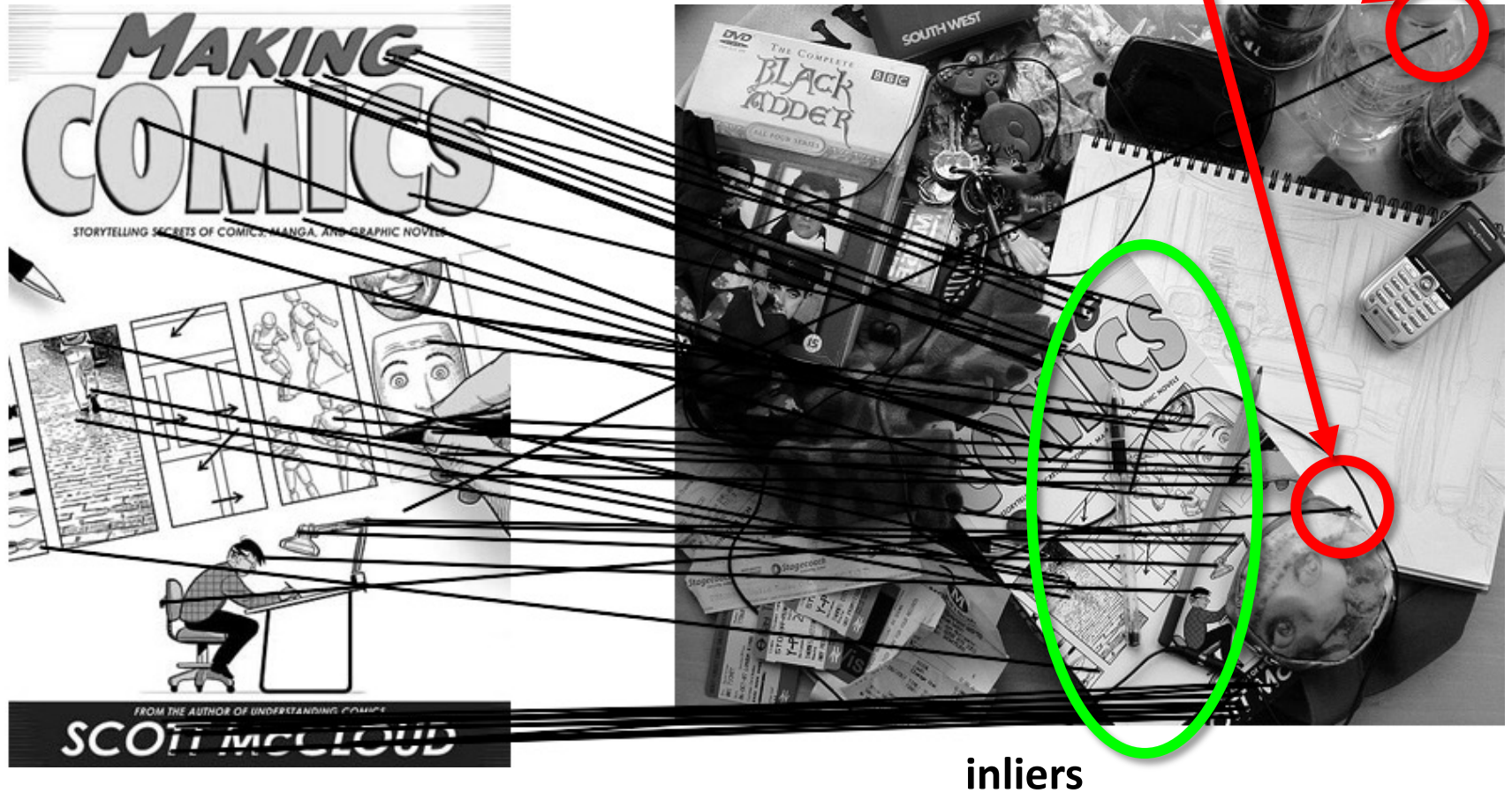
$\mathbf{h}$
 9

$\mathbf{0}$
 $2n$

Defines a least squares problem: minimize $\|\mathbf{A}\mathbf{h} - \mathbf{0}\|^2$

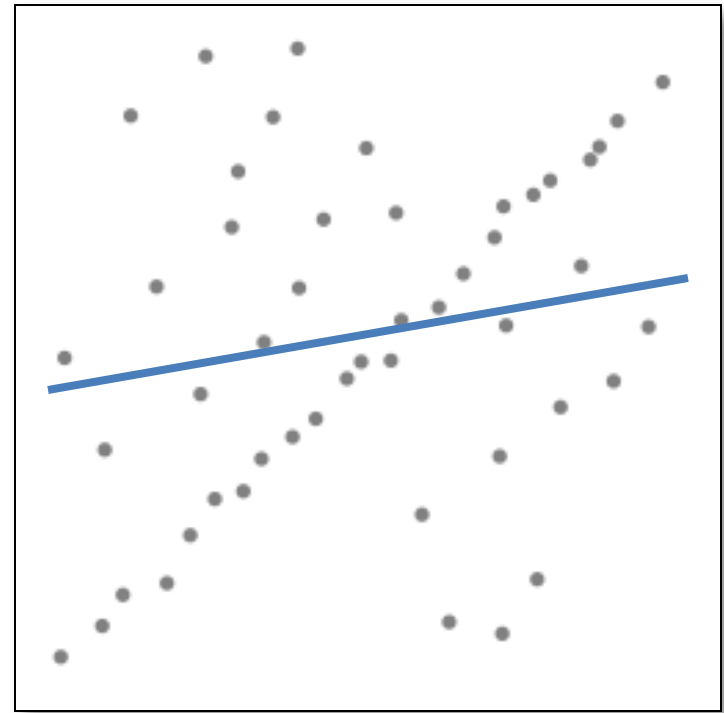
- Since $\mathbf{h}$ is defined up to scale, solve for unit vector $\hat{\mathbf{h}}$
- Solution: $\hat{\mathbf{h}}$ = eigenvector of $\mathbf{A}^T \mathbf{A}$ with smallest eigenvalue
- Works with 4 or more points

Outliers



Robustness

- Let's consider a simpler example... linear regression

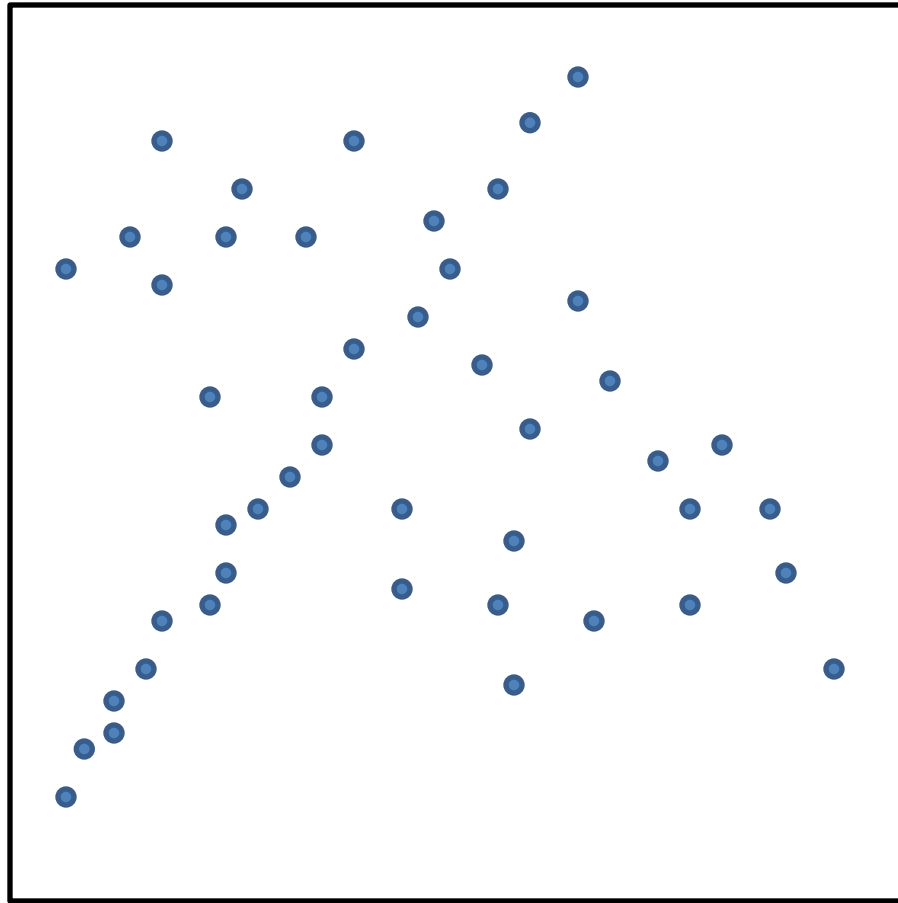


Problem: Fit a line to these datapoints

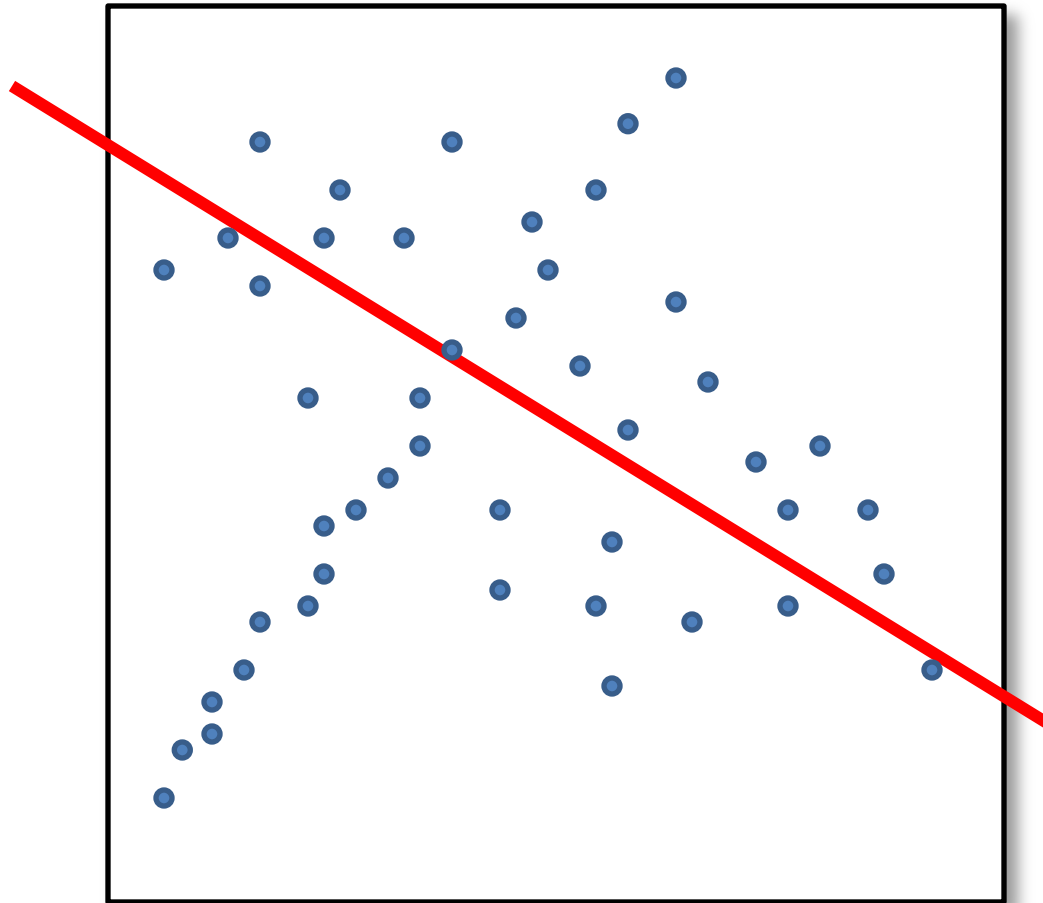
Least squares fit

- How can we fix this?

RANSAC

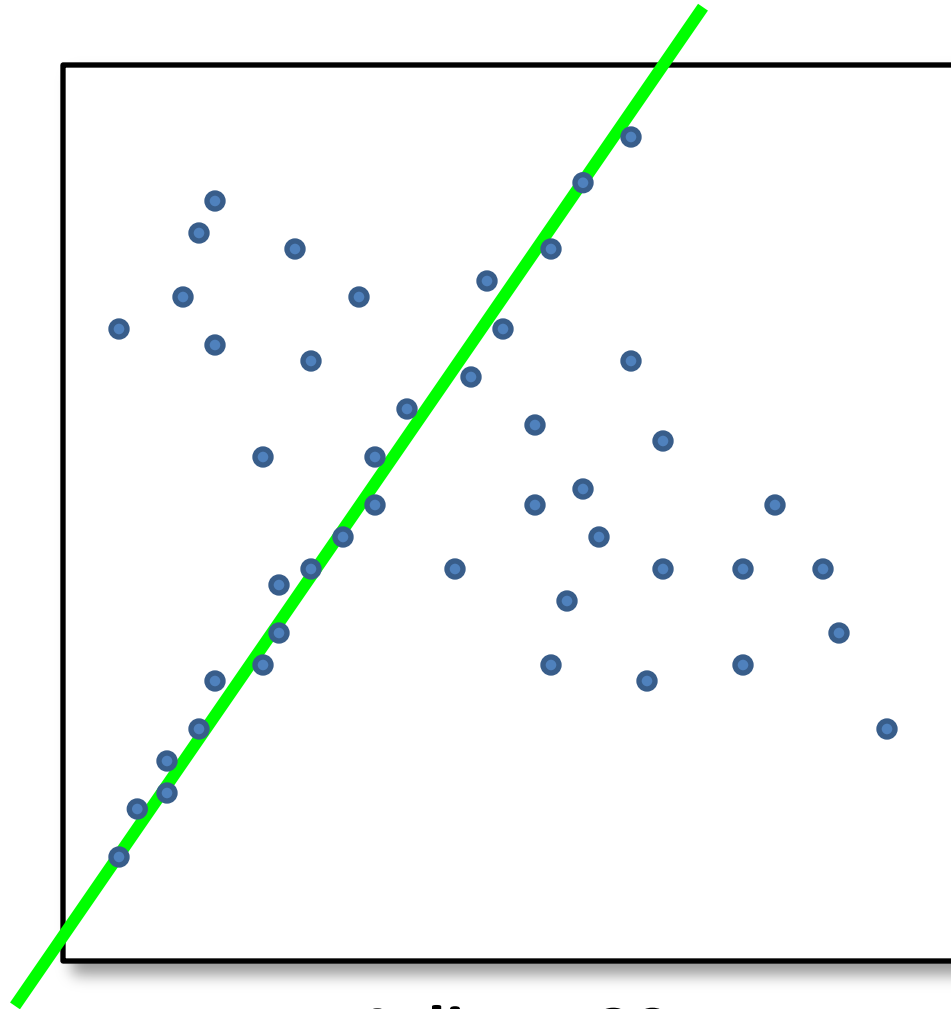


RANSAC



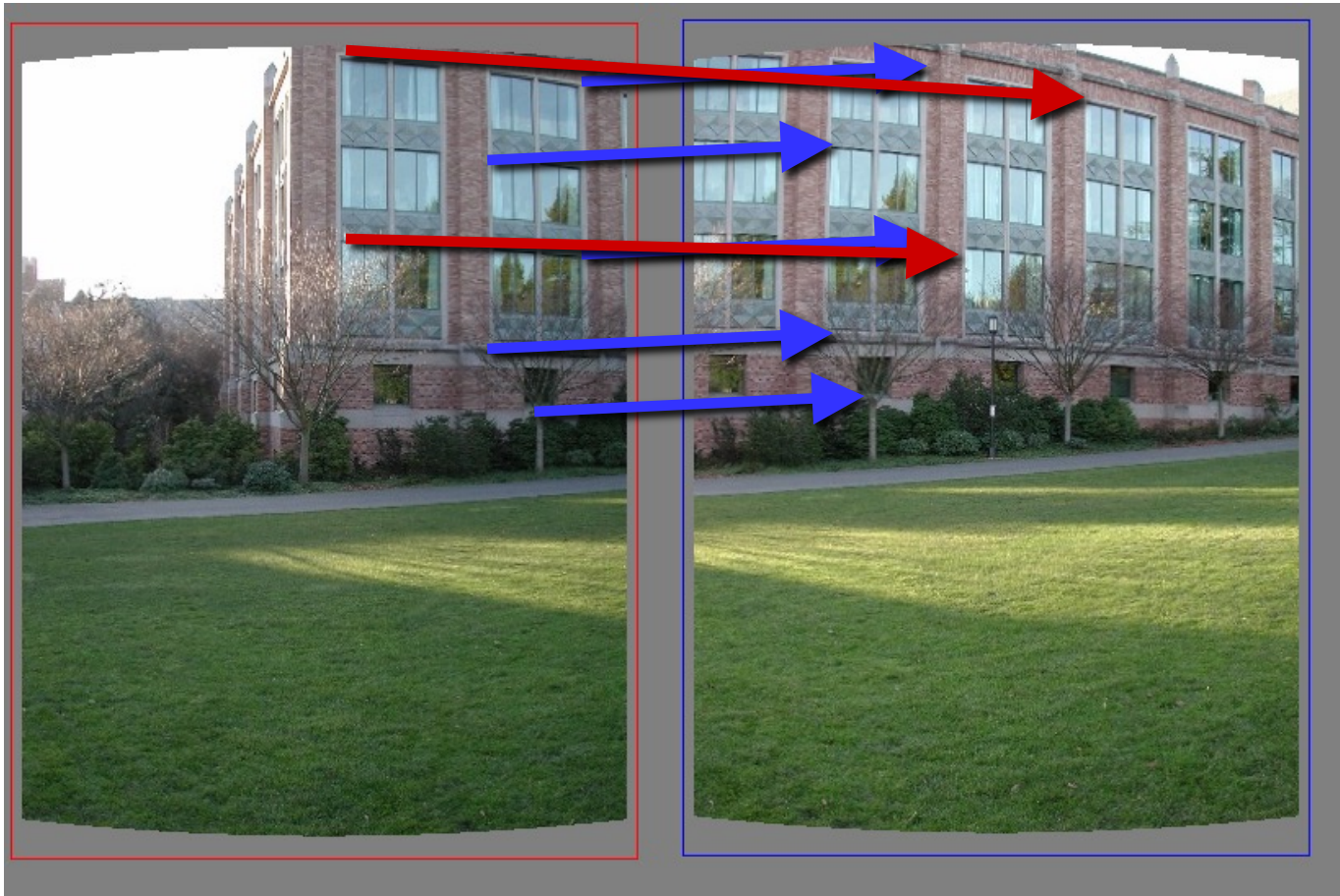
Inliers: 3

RANSAC

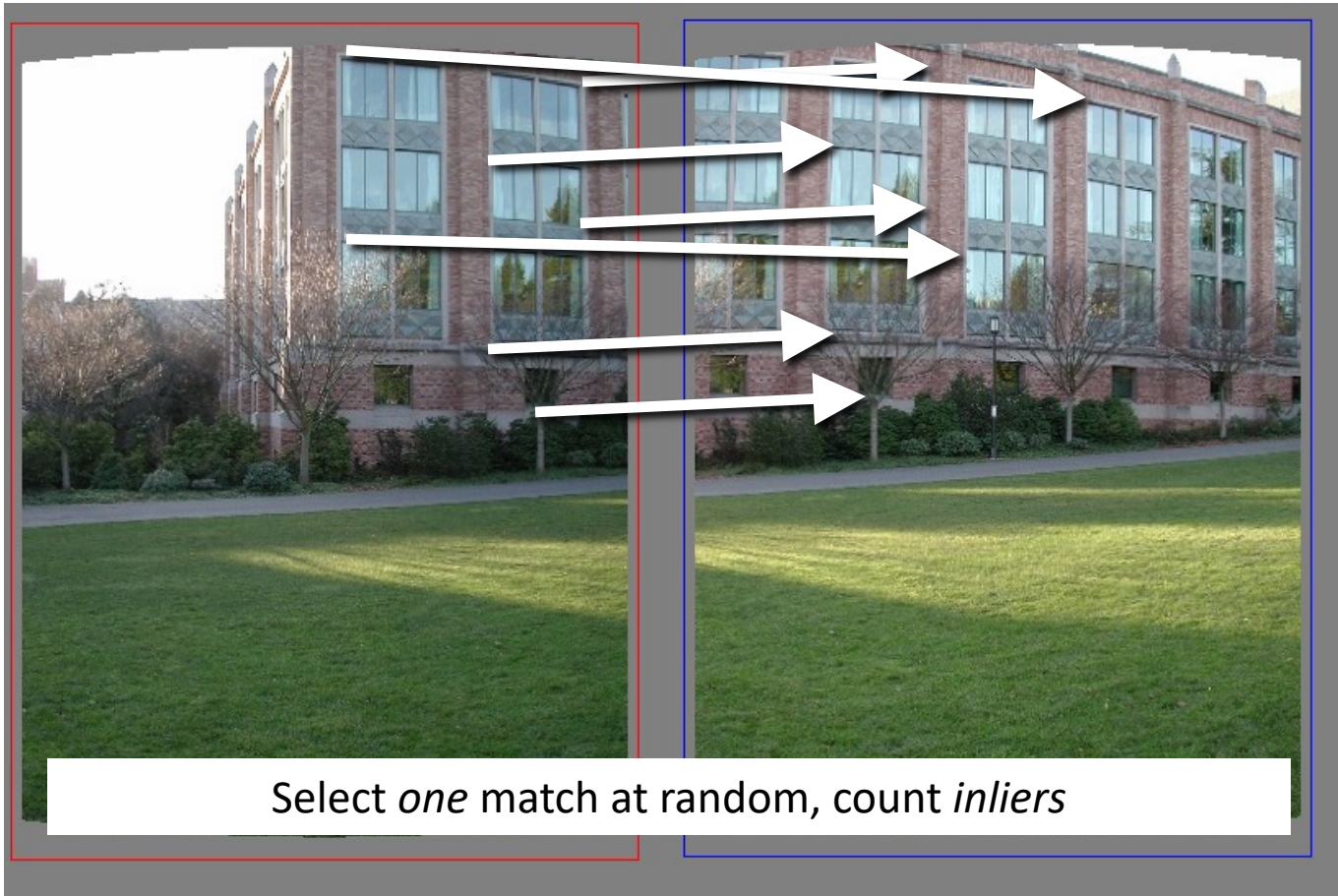


Inliers: 20

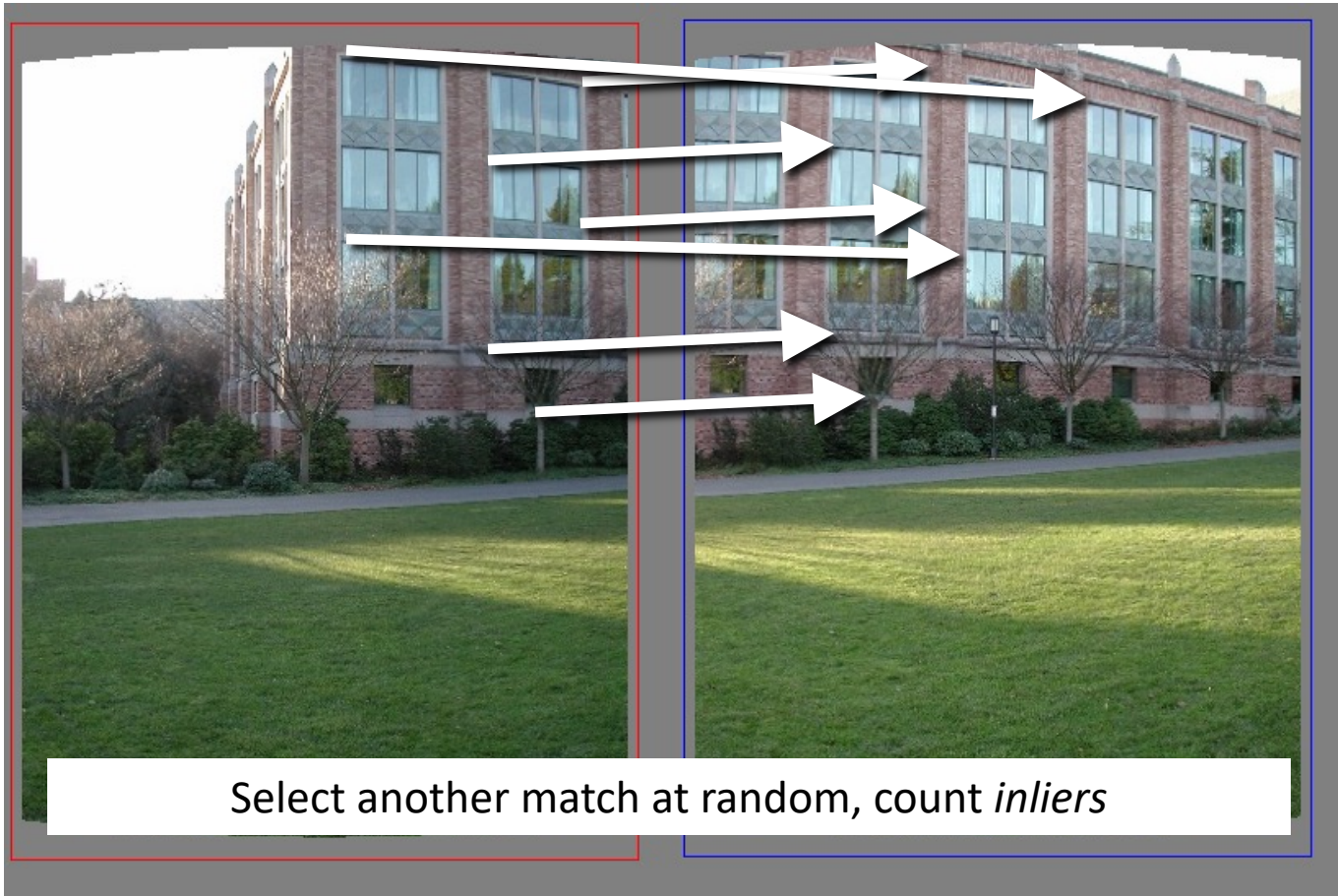
RANSAC for Translation



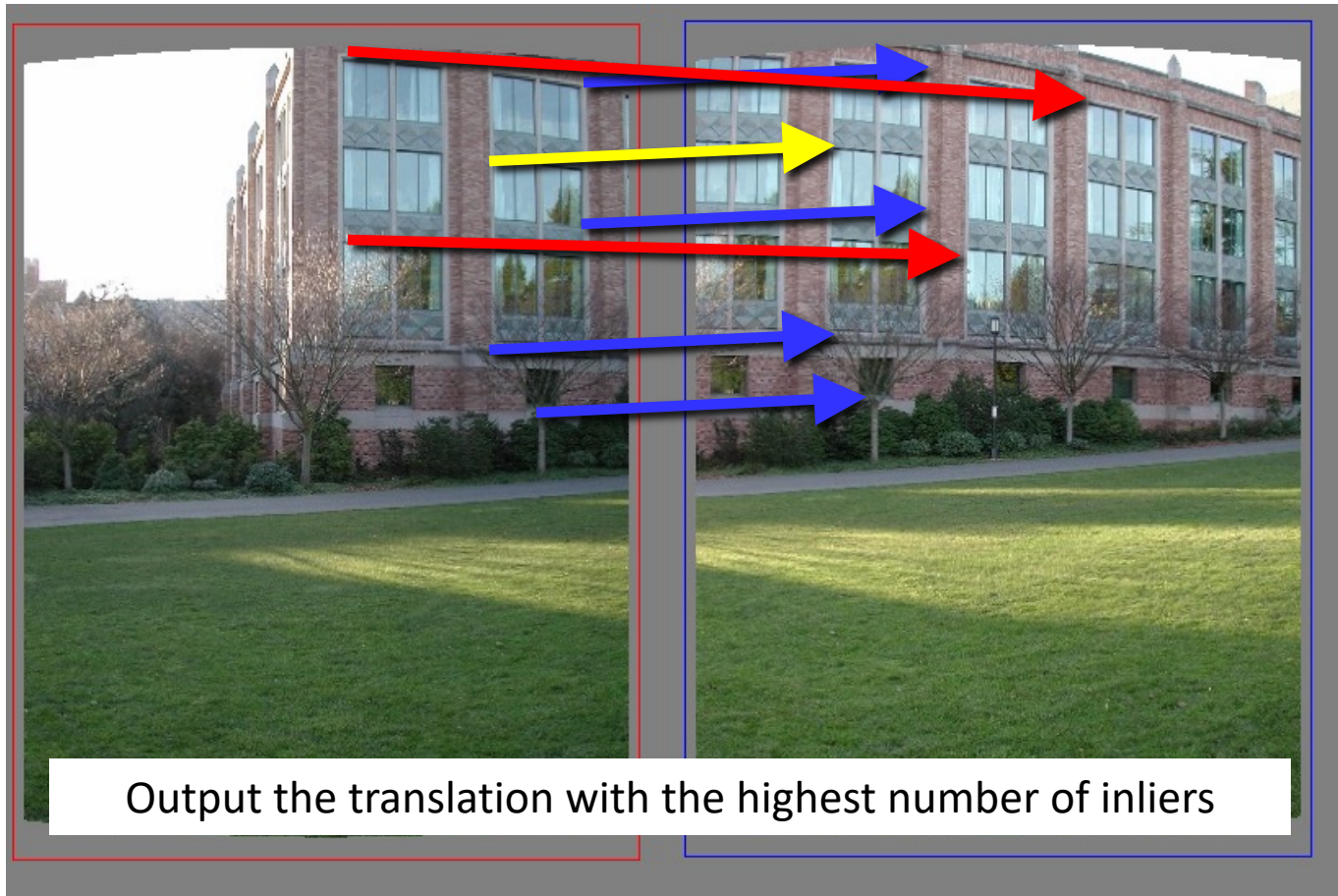
RANSAC for Translation



RANSAC for Translation



RANSAC for Translation



RANSAC

- Idea:
 - All the inliers will agree with each other on the translation vector;
 - The outliers will disagree with each other
 - RANSAC only has guarantees if there are $< 50\%$ outliers
 - “All good matches are alike; every bad match is bad in its own way.”
 - Tolstoy via Alyosha Efros

RANSAC

- General version:
 1. Randomly choose s samples
 - Typically s = minimum sample size that lets you fit a model
 2. Fit a model (e.g., transformation matrix) to those samples
 3. Count the number of inliers that approximately fit the model
 4. Repeat N times
 5. Choose the model that has the largest set of inliers

Final step: least squares fit

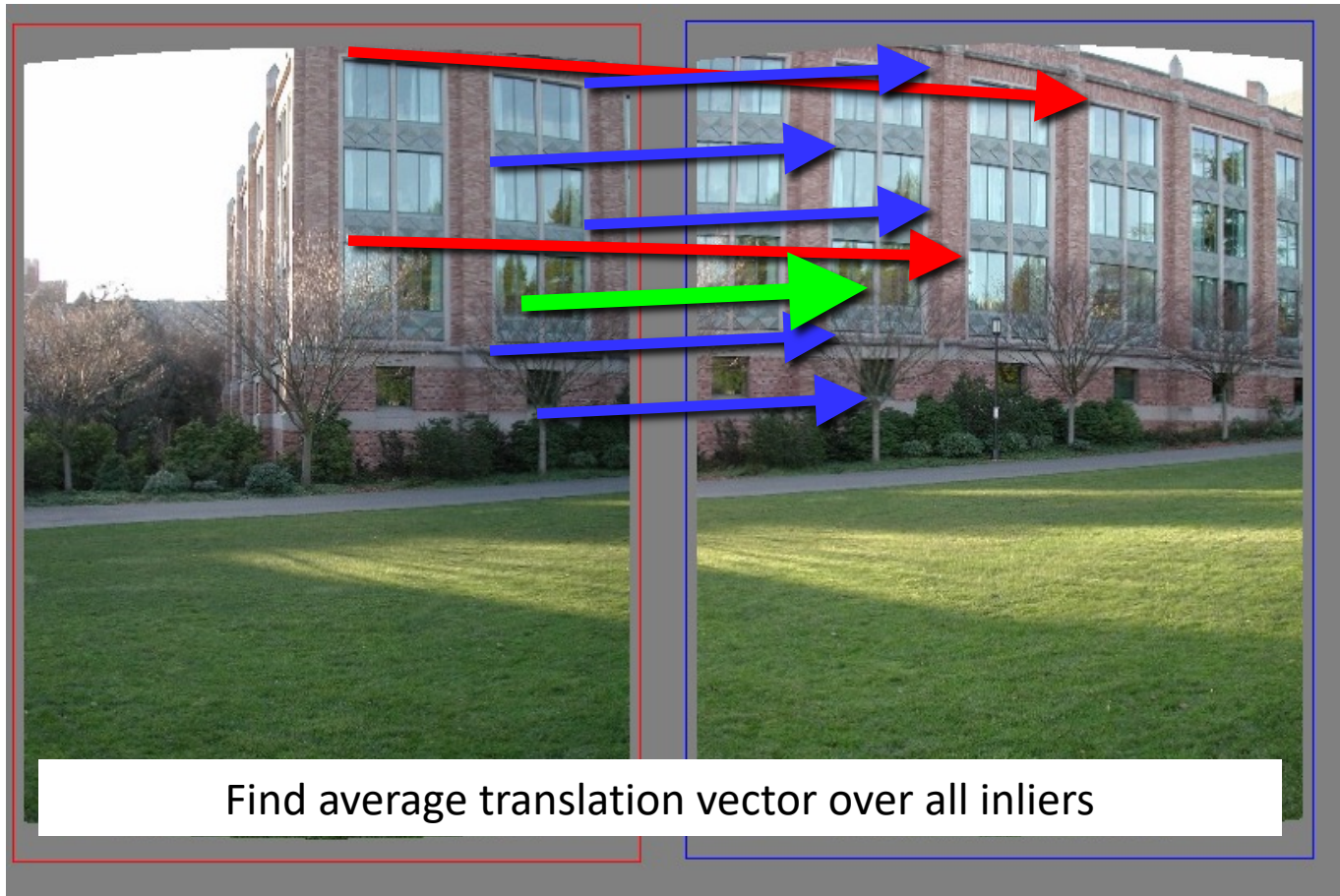
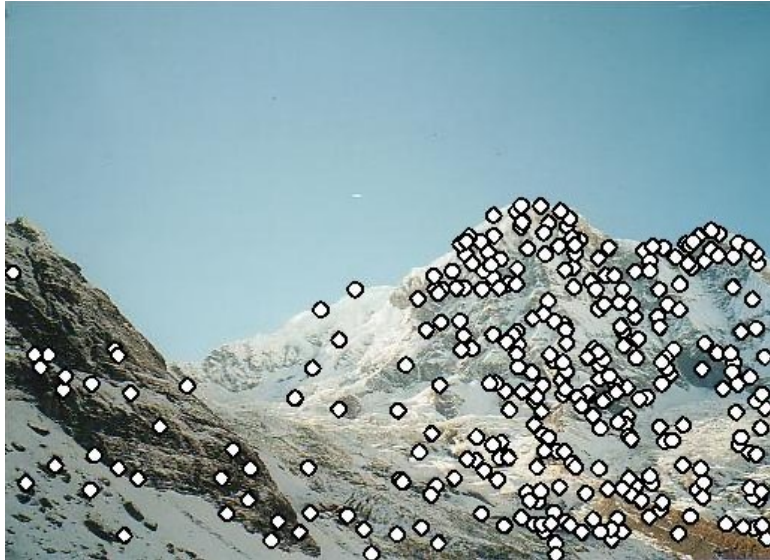


Image Stitching



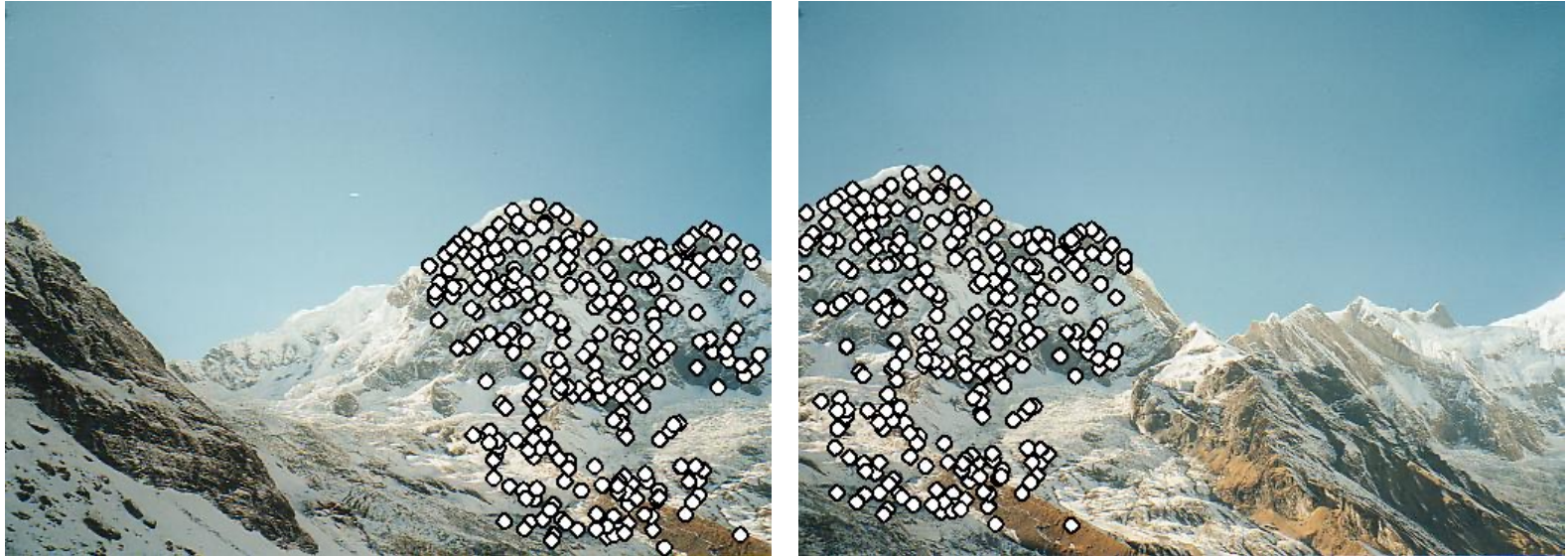
Input images

Image Stitching



Feature matching

Image Stitching



Compute transformation matrix with RANSAC

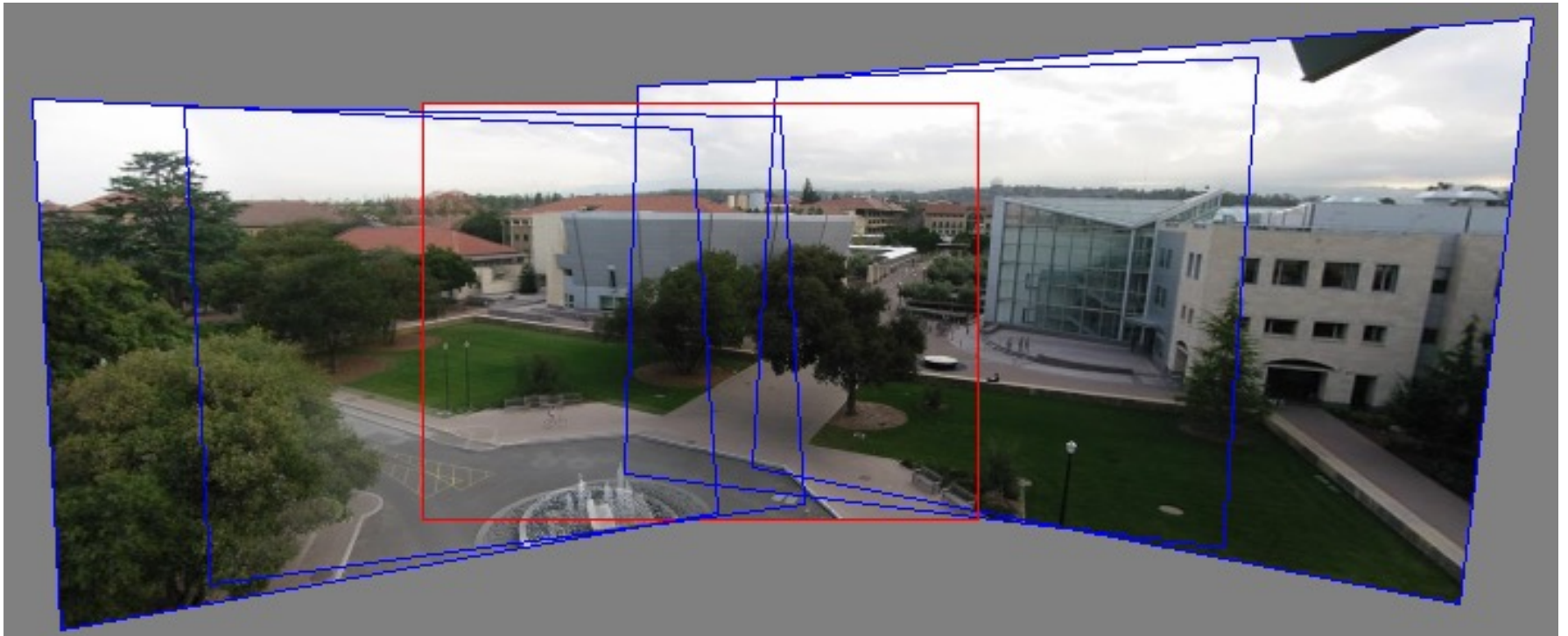
Image Stitching



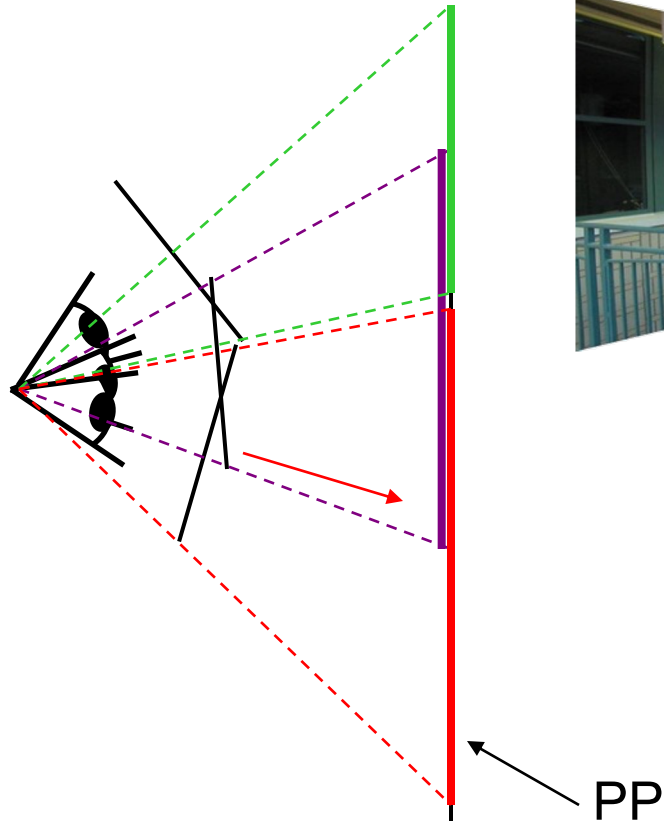
Fix image 1 and warp image 2

Panoramas

- Now we know how to create panoramas!
1) Warp all images to a reference image; 2) merge them

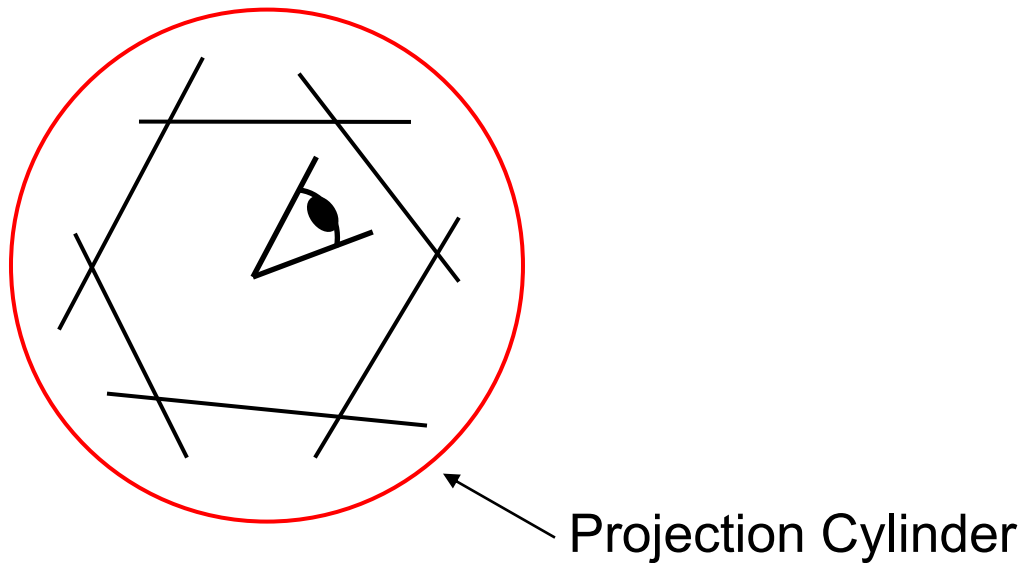


Do we have to project onto a plane?

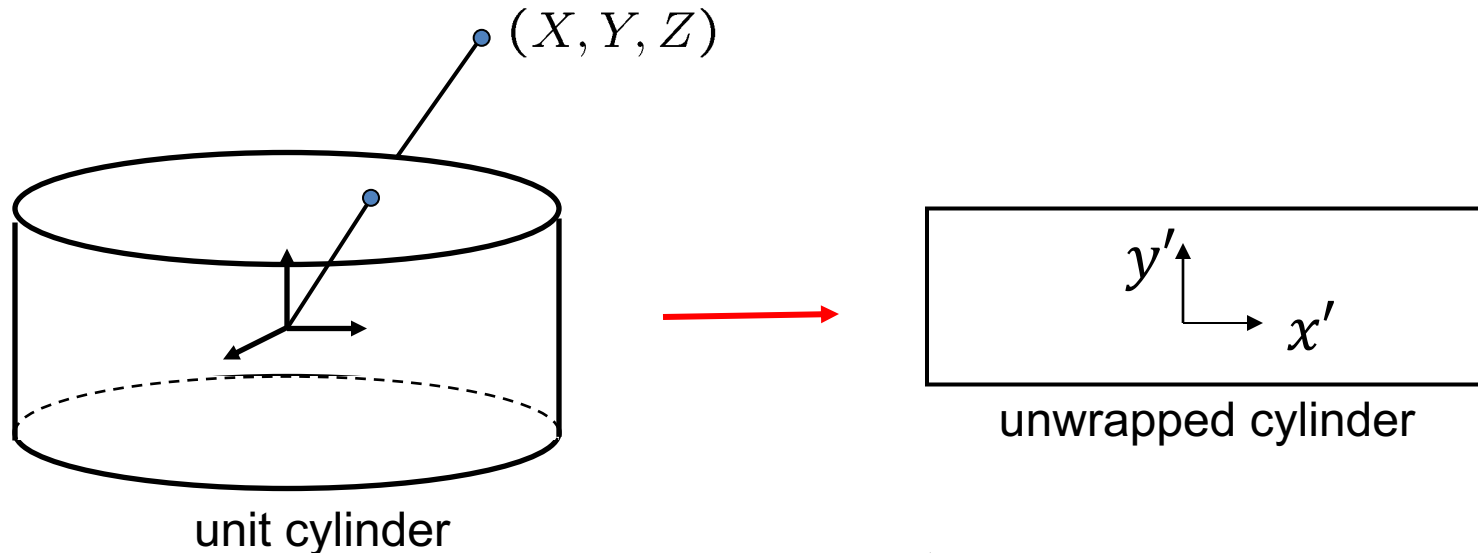


Full Panoramas

- What if you want a 360° field of view?



Cylindrical projection

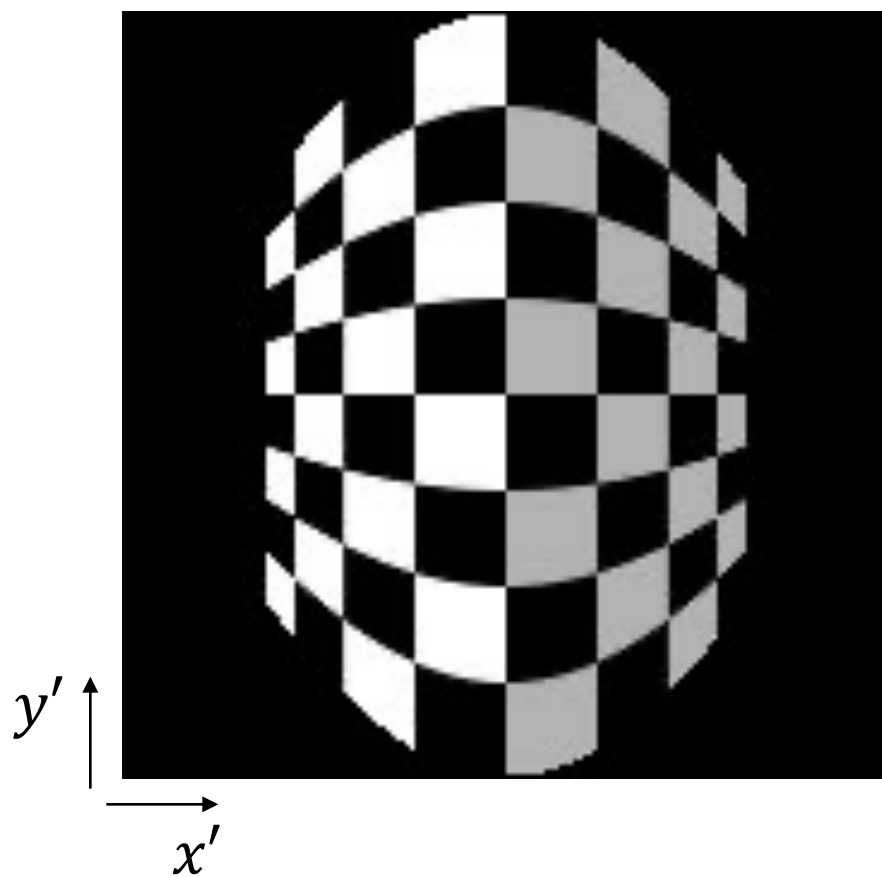
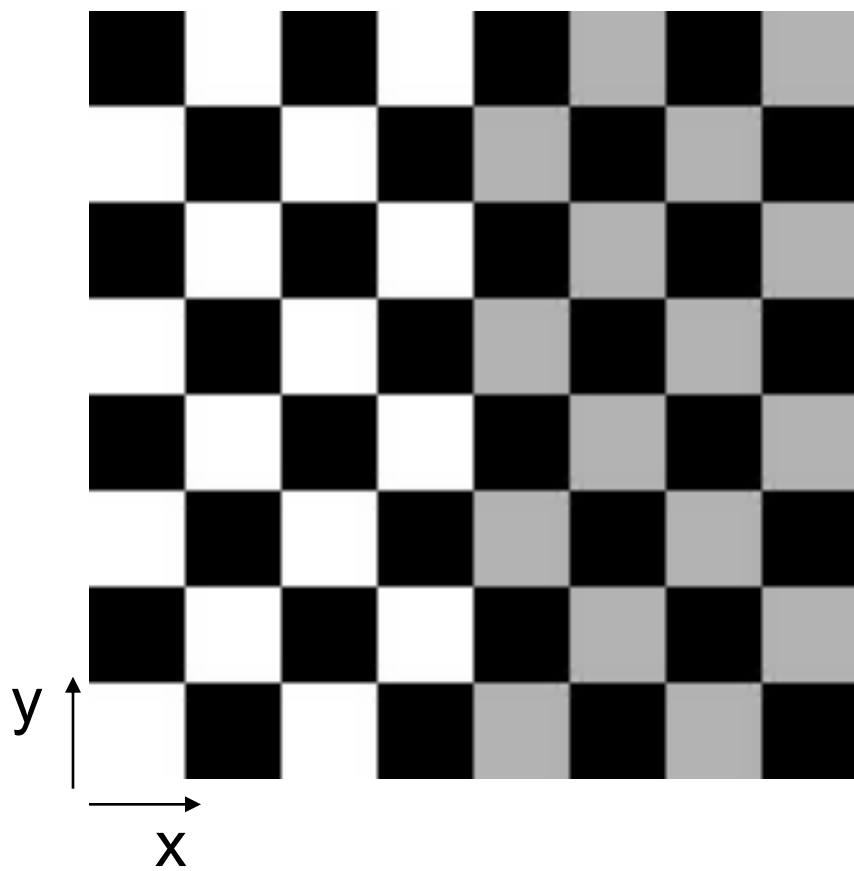


$$x' = r \tan^{-1} \left(\frac{x}{f} \right)$$

$$y' = \frac{ry}{\sqrt{x^2 + f^2}}$$

(x', y') are cylindrical coordinates, (x, y) are image coordinates (origin at image center), r is the radius of cylinder, f is the focal length

Cylindrical projection



Cylindrical panoramas



- Steps
 - Reproject each image onto a cylinder
 - Blend

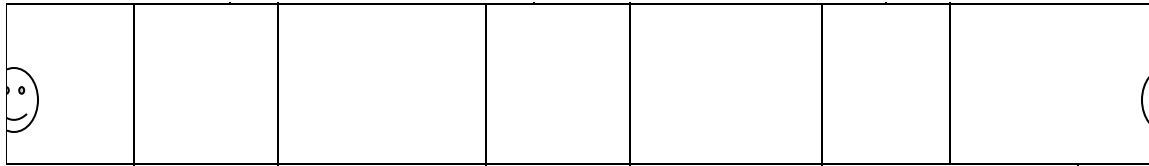
Cylindrical image stitching



How to compute the transformation on cylinder?

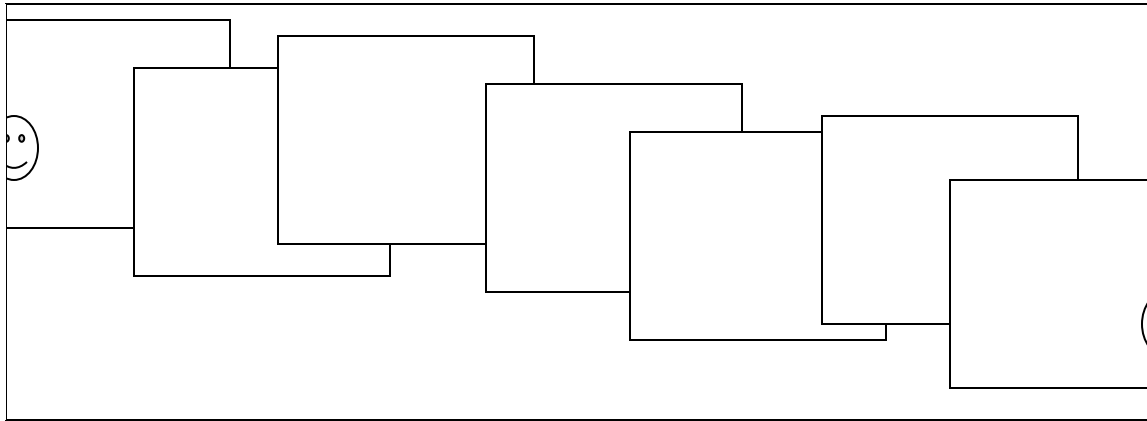
- A rotation of the camera is a **translation** of the cylinder!

Assembling the panorama



- Stitch pairs together, blend, then crop

Problem: Drift



- small (vertical) errors accumulate over time
- apply correction so that sum = 0 (for 360° pan.)

Full-view (360°) panoramas







Questions?